

OBJECT ORIENTED PROGRAMMING

Assignment 3

Topic: Inheritance, Polymorphism Association, Aggregation, Composition in C++

Duration: 1 Week

Total Marks: 15

Submission Deadline: 11-01-2025

CLO	CLO Statement	Blooms Taxonomy Level	PLO
3	Apply object-oriented programming principles to implement small and medium problems	C3	PLO-3

Task 1: Online Shopping Platform

An online store sells multiple types of products. Each product has some kind of category. A shopping cart contains products but does not own them permanently. User can add item into the cart and remove item from cart.

Classes & Members:

- **Product**
 - o Attributes: productId, price, name
 - o Functions: getPrice(), applyDiscount()
- **Category**
 - o Attributes: categoryId, Name, Description
 - o Functions: applyDiscount()
- **Electronics / Clothing (Derived)**
 - o Attributes: warrantyPeriod / size
 - o Functions: applyDiscount()
- **ShoppingCart**
 - o Attributes: listOfProducts
 - o Functions: addProduct(), removeProduct(), calculateTotal()

In main function user can view any product or products of any category, Apply Discount on single product of any Category products. Also, if cart total amount exceeds from 10,000 rupees it automatically eligible for 15% discount so deduct that amount from the Cart Total Amount. Along with that user can view his shopping cart when it required.

Task 2: Hospital Management System

Explanation:

Doctors and patients interact temporarily. Medical records belong strictly to patients.

Classes & Members:

- **Doctor**

- o Attributes: doctorId, specialization
 - o Functions: treatPatient()
- **Patient**
 - o Attributes: patientId, name
 - o Functions: consultDoctor()
- **MedicalRecord (Composed)**
 - o Attributes: diagnosis, treatment
 - o Functions: updateRecord()

Task3: Media Player Application

Explanation:

Different media types are played using the same interface.

Classes & Members:

- **MediaFile (Base Class)**
 - o Attributes: fileName, duration
 - o Functions: play() (virtual)
- **AudioFile / VideoFile**
 - o Attributes: bitrate / resolution
 - o Functions: play() (override)

Task4: Ride-Sharing Application

Explanation:

Fare calculation depends on vehicle type.

Classes & Members:

- **Vehicle (Base Class)**
 - o Attributes: vehicleNumber
 - o Functions: calculateFare()
- **Car / Bike**
 - o Attributes: seatCount
 - o Functions: calculateFare() (override)
- **Driver**
 - o Attributes: driverId
 - o Functions: acceptRide()

Task 5: Smart Home Automation

Explanation:

A smart home controls multiple devices with common commands.

Classes & Members:

- **Device (Base Class)**
 - o Attributes: deviceId
 - o Functions: turnOn(), turnOff()
- **Light / Thermostat**
 - o Attributes: brightness / temperature
 - o Functions: turnOn() (override)
- **Home**
 - o Attributes: devices
 - o Functions: controlAllDevices()